

ETHAN TUROK

[[Website](#), [Github](#), [Email](#), [Google Scholar](#)]

SUMMARY: Research scientist who trains state-of-the-art LLMs, publishes in top-tier ML conferences (ICML/ICLR), and builds systems used by Fortune 500 companies.

EDUCATION

Weizmann Institute of Science	Overseas
M.Sc. Computer Science & Mathematics <i>Full Scholarship</i>	Expected 2026
Columbia University, School of Engineering and Applied Sciences	New York, NY
B.Sc. Computer Science & Applied Mathematics GPA: 3.93 <i>Magna cum Laude</i>	May 2023

EXPERIENCE

MosaicML x Databricks	New York, NY
<i>AI Research Scientist Advisor: Jonathan Frankel</i>	2023-24

- Trained [DBRX](#), a state-of-the-art LLM (132B MoE)
- Integrate **tensor parallelism** for 8x memory savings, linear scaling across **2,000+ GPUs**
- [Developed](#) **structured generation** for LLMs to use tools; used by **1,000+ customers** at Fortune 500 companies
- Pretrain and finetuned **1M-70B+ param models** on petabytes of internet data and custom synthetic datasets
- Lead researcher on data tagging: augment dataset to improve LLM coding by 50% **without** any extra finetuning

Pe'er Lab, Columbia University	New York, NY
<i>Deep Learning Researcher Advisor: Itsik Pe'er</i>	2022-23

- Trained **200+** deep learning models on genetic data in *hyperbolic* space to improve disease prediction [[preprint](#)]
- Showed *hyperbolic* embeddings enable *efficient* representation learning: **5x lower MSE** for predicting genetic similarity and **2x faster** for approximate nearest neighbor search [[preprint](#)]
- Developed *efficient* random forest algorithm in *hyperbolic* space: up to **60x faster**, **25% more accurate** [[arXiv](#)]

Theoretical Computer Scientist, Columbia University	New York, NY
<i>TCS Researcher Advisors: Josh Alman, Rachel Cummings</i>	2022-23

- Developed **faster algorithms** for dynamic programming problems via fine-grained complexity reductions [[arXiv](#)]
- Built machine learning algorithms that **protect user's privacy** via differential privacy [[arXiv](#)]

StudBud	New York, NY
<i>Startup Co-founder</i>	2020-22

- Co-founded a [startup](#) to connect students during COVID; won the [Columbia Innovation, Enhancement Award](#)
- Matched **1300+ students** at Columbia into ideal study groups with hierarchical clustering algorithm

AWARDS

Lambda Labs Research Grant Awarded GPU compute for research	2025
Monday.com AI Hackathon Finalist Built LLM chatbot and won \$3000 among 1.5k+ participants	2023
Columbia Community Building Award Excellence in teaching and fostering community in CS department	2022
Columbia Innovation & Enhancement Award Built ML tool to match students into study groups during COVID	2021
Columbia University Dean's List All available semesters	2020-23

EXTRACURRICULAR ACTIVITIES

Open Source Contributor: [transformers](#), [tinygrad](#), [outlines](#), [Gorilla](#), [MegaBlocks](#)

Competitive Programming: Study advanced algorithms, data structures with Columbia ICPC team

VC Consultant: Provide technical diligence for VCs on cutting edge tech companies

PAPERS

1. **Differential Privacy Meets Imbalanced Learning: Methods and Empirical Insights.** Lucas Rosenblatt, Yuliia Lut, Ethan Turok, Marco Avella Medina, Rachel Cummings. *International Conference in Machine Learning (ICML)*, 2025. [[arXiv](#)]
2. **Beyond the Leaderboard: Unpacking Function Calling Evaluation.** Kartik Sreenivasan, Jeffrey Chen, Pallavi Koppol, Ethan Turok, Bay Foley-Cox, Asfandiyar Qureshi and Sam Havens. [[blog-post](#)]
3. **Introducing DBRX: A New State-of-the-Art Open LLM.** The Mosaic Research Team. [[blog-post](#)]
4. **Tensors Ranks and the Fine-Grained Complexity of Dynamic Programming.** Josh Alman, Ethan Turok, Hantao Yu, Hengzhi Zhang. *Innovations in Theoretical Computer Science (ITCS)* 2024. [[arXiv](#)]
5. **Fast hyperboloid decision tree algorithms.** Philippe Chlenski, Ethan Turok, Antonio Moretti, Itsik Pe'er. *International Conference on Learning Representations (ICLR)*, 2024. [[arXiv](#)]

SKILLS

Programming Languages	Python, Java, C++, LaTeX
Technologies/Packages	Git, Bash; Numpy, Scikit-Learn, Scipy; Pandas, Spark; PyTorch, W&B, HuggingFace
Interests	Rock climbing, snowboarding, jazz concerts in the city